



A real Hoverboard?

This super cool one-wheeler rides like a hoverboard. Check it out:
<http://dgit.in/1PQUabo>

All Rounder

Vivint has developed a new home automation system with a slight addition:
<http://dgit.in/1Gmcybv>



Build your own solar charger

DO your bit for the environment by building your own solar charger. It's much easier than you'd think.

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Sunlight is free and at the very least, it can be used to charge your phones. Follow this guide and in five minutes you'll have a dirt cheap solar charger to play around with.

A solar cell converts sunlight to DC



A sampling of all the stuff you'll need to build the charger

voltage and since that's all a battery needs to charge itself, it's the perfect "green" solution for charging your batteries.

Now for some theory. A normal mobile phone charger provides around 5V and 1A of current and this is enough to charge most phones at a reasonably quick rate. You will, of course, find certain chargers that charge at 2.1A or more, and these are usually required to charge more powerful devices that require larger batteries.

The point here is that the required voltage remains the same at 5V, but the amount of current varies depending on what you're charging. Up to a certain point, the more current your circuit produces, the faster your phone will charge and vice versa.

Most solar cells are rated for only about 1V and 20mA, which is nowhere near enough. You have two options here, either purchase a more powerful panel (₹400 for an 8V, 500mA panel). Alternately, if you have one of those solar garden lamp or other such contraption, you can recycle the solar cells for your charger.

What you need:

- A bunch of solar cells that can provide at least 5V (either in series or in parallel)
- Some wire
- Insulation tape
- A USB car charger or old USB cable
- A multimeter (optional, but it's best to have one around)
- Soldering gun + solder (optional)
- Diode (optional if you know what you're doing or are using a car charger)

Assembly

Step 1

First you need to determine the total current and voltage that each cell can provide. Once you've noted that down, you need to remember that you need to provide between 4-6V before your phone will start charging.

The amount of current only determines how fast your device will charge but a lower voltage will mean that your phone won't even start charging.

Once you figured out the voltages